Project Title: **Predicting Student Placements Using Random Forest Classifier**

Dataset Name: **student_dataset_10000_rows.csv (Kaggle Student Performance & Placement Dataset)**

Problem Statement:

In this project, I wanted to build a machine learning model using the Random Forest algorithm to figure out if a student gets placed or not. I used a dataset with student details like study hours, attendance, sleep hours, internet usage, assignments completed, previous scores, and exam scores to see how these factors affect placement results.

Data Preprocessing Steps

Checking the Data: I loaded the dataset which has 10,000 rows and 8 columns, and verified there were no missing values.

Label Encoding: Since placement_status was in text format (Placed and Not Placed), I converted it into numbers (1 and 0) using Label Encoder.

Separating Features: I separated the input features (X) from the target column (y).

Train-Test Split: I split the data into 80% for training the model and 20% for testing it.

Model Used

Random Forest Classifier: I chose the Random Forest model with 100 decision trees (n_estimators=100) because it combines multiple decision trees to give better accuracy and prevents overfitting.

Evaluation Metrics

Since this is a classification task, I tested the model on the test data using these 5 metrics:

Accuracy: 1.0000 (100% correct predictions on the test set)

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Confusion Matrix: The matrix showed 0ne misclassifications on the test data.

Key Findings

Main Predictors: exam_score and study_hours were the most important features in deciding whether a student got placed.

Other Factors: Features like assignment completion and attendance also had a good impact on student performance.

Model Performance: The Random Forest algorithm worked really well on this dataset without any errors.

Conclusion: Building this Random Forest model helped me understand how ensemble learning works on real data. The model learned the patterns in student performance easily and predicted placements with high accuracy.

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